

Impact of behavioral biases on investment decisions and the moderation effect of financial literacy; an evidence of Pakistan

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ARTICLE INFO

Keywords:

Behavioral finance

Biases

Investment decision-making

Financial literacy

Individual investors

ABSTRACT

This study offers empirical insights into investor behavior and its correlation with various behavioral biases in the context of investment decisions in the Pakistan Stock Exchange (PSX). Data was collected through a structured questionnaire from 261 individual investors in Pakistan. The study employs hierarchical regression analysis to test the hypothesis. It considers several behavioral biases, and statistically, anchoring and adjustment, overconfidence, and herding show a significant impact. The study uses financial literacy to examine its moderating effects on these biases, and the result suggests that it significantly influences behavioral biases related to investment decisions. The results underscore the unique investment behaviors in emerging markets, contrasting with established norms in well-developed financial markets. These findings can inform policymakers and stock market authorities about investor decision-making in emerging economies.

1. Introduction

Financial decisions are essential and necessary in a family's financial and personal wealth management (Sahi et al., 2013). The conventional finance theories focusing on utility maximization assume that markets are efficient and investors are rational in their decision-making. In efficient markets, information reaches the market quickly and equally. For rational decision-making, investors collect and process this information. Based on the objective information and their attitude, investors make investment decisions. However, in traditional finance, there is a lack of consensus on the efficiency of the financial markets and rational decision-making of investors. The introduction of behavioral finance further challenges the efficiency of markets and rationality in decision-making. In response to the rationality of "economic man" Simon (1955), in the light of psychological theories, initially introduced the impact of aspirations, where emotions dominate the limited knowledge or intellect in decision-making.

Extending a similar concept, Tversky and Kahneman (1974) proposed prospect theory, assuming that investors' primary decisions are more influenced by psychological priorities than rationality. Investors apply heuristics or shortcuts to overcome their limits of knowledge and

rationality. Heuristics describes valuable and vital techniques for discovering answers to problems that cannot be addressed using logic or probability theory (Bazerman & Moore, 2012). As per Yamini (2020), heuristics can be helpful in this regard; they can cause errors and systemic distortions from rational approaches. Understanding how physiological and behavioral biases influence individual speculators' judgments and the complicated decision-making process of picking investments may enhance decision-making ability, reduce decision-making errors, and improve individual speculators' performance (Yao-wen et al., 2015).

According to Shefrin (2001), from a behavioral finance perspective, psychological factors significantly influence investors' rational decision-making. Therefore, the behavioral aspect that impacts an individual's investing choice is worth exploring, particularly investors' behavioral biases (Quaicoe & Eleke-Aboagye, 2021). Hence, by observing the attitude and behavior of investors in financial markets in recent decades, behavioral finance integrated the theories of sociology and psychology with finance. In traditional finance, people, organizations, and perhaps even markets are believed to be rational (Baker & Filbeck, 2013). Individuals consider all available facts, risk avoiders, and logic and choose lower risky securities at the same gain level (Arora & Kumari, 2015;

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<https://doi.org/10.1016/j.actpsy.2024.104303>

Received 1 September 2023; Received in revised form 24 April 2024; Accepted 7 May 2024

Available online 25 May 2024

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Shah, Ahmad, & Mahmood, 2018). They adopt theoretical concepts and assumptions of traditional finance to predict and measure risks before investing in different securities (Mushinada & Veluri, 2019). However, multiple studies (Adil, Singh, & Ansari, 2021; Ahmad, 2021; Özen & Ersoy, 2019) show that investors behave irrationally and markets are inefficient (Ahmad & Shah, 2020; Bashir et al., 2021; Shah, Ahmad, & Mahmood, 2018). Similarly, Shefrin (2001) argued that psychological (behavioral) factors negatively influence investors' investment decisions. In financial markets, many investors do not possess sufficient knowledge and skills, and when making investment decisions, they use their subjective intuition and emotions (Antony & Joseph, 2017). Due to individual attitudes and behavioral biases, investors make trading mistakes, trade aggressively, purchase stocks without evaluating their intrinsic worth, invest in a stock their peers are purchasing, make financial market selections based on prior records, and hold losing investments while selling profitable ones.

According to behavioral finance, investors' psychological and emotional behavior deviates from rational behavior (Bakar & Yi, 2016; Yoong & Ferreira, 2013) due to mental decline that leads to bad investing decisions. Behavioral finance highlights various inefficiencies in financial markets by examining the behaviors of investors in various market conditions contradictory to the traditional norms (Yoong & Ferreira, 2013) and results in illogical decisions made by investors (Özen & Ersoy, 2019). Behavioral finance research has recently uncovered a few behavioral biases that impact investment decisions. Researchers found that everyone has cognitive errors that influence investment decisions (De Bondt et al., 2013; Hussain et al., 2013; Shah, Ahmad, & Mahmood, 2018), leading to adverse investing outcomes and poor investment performance (Bashir et al., 2013). In this regard, Kahneman and Tversky (1979) identified psychological traits, age, gender, and education level, while (Metawa, Hassan, Metawa, & Safa, 2019; Shefrin, 2002; Statman et al., 2006) highlighted emotional factors like anxiety, selfishness, and overconfidence that can substantially influence investment decisions.

Apart from psychological traits, existing literature highlights the significance of financial knowledge/literacy in investment decision-making. The studies by Takeda et al. (2013), Grohmann and Menkhoff (2015), Adil, Singh, & Ansari, 2021, Khawar and Sarwar (2021), and Khan et al. (2024) reported that investment decisions are the outcome of investors' awareness and knowledge about financial products. Investors' financial literacy becomes more important in global economic integration and technological development, resulting in financial advancement and easy and quick access to financial products and services (See Herawati et al., 2018; Khan & Al-Harby, 2022). Studies like Klapper et al. (2013) concluded that investors' attitudes and limited financial knowledge contributed to the 2007–2008 financial crises. Creating risky investment opportunities, financial institutions attracted significant amounts of investments. The investors invested in such investments for higher returns without analyzing the risk and default factor of those investment opportunities. Post-crisis research highlighted that low financial literacy resulted in irrational or harmful financial decisions Sezer and Demir (2015) and Son and Park (2019). They proposed that financial literacy can minimize the problems that arise from investors' low levels of financial information, which could result in a higher risk of making poor investment selections. Extending a similar argument, Gigerenzer and Gaissmaier (2011) examine the relationship between financial literacy and heuristic variables. They found that combining the heuristics with financial knowledge can reduce the negative impact of heuristic variables. Individuals change their behavior depending on the expertise and information gathered and invested based on their knowledge and financial experience. A knowledgeable investor may overcome their biases and make sound financial decisions.

Therefore, several studies examine the role of financial literacy in making smart investing decisions (Adil, Singh, & Ansari, 2021; Baker, Kumar, Goyal, & Gaur, 2019; Rasool & Ullah, 2020). Hence, considering the importance and role of the abovementioned factors, the study's

primary goals are (1) to investigate whether behavioral biases influence individual investors in Pakistan. (2) to determine the influence of behavioral biases on investor decision-making. (3) to determine the role of financial literacy in the link between behavioral biases and investment decision-making.

The empirical literature on behavioral finance has overlooked exploring the behavior of investors in the emerging market of Pakistan. Hence, this study contributes to the existing body of knowledge by filling this gap by adding empirical evidence on investors' behavioral biases and investment decision-making in Pakistani emerging financial markets. Moreover, the study also uses representativeness and anchoring bias as independent variables along with financial literacy as a moderating variable to comprehensively examine the investor's behavior and the impact of these on their financial decision-making, the use of which in current literature is limited. Based on the mentioned research gap, the following are the contributions of this study in behavioral finance. Firstly, this study fills the research gap, providing empirical evidence on how behavioral biases influence investors' investment decisions in the Pakistani financial markets. It is observed that behavioral biases, anchoring and adjustment, overconfidence, and herding show a statistically significant impact on investors' decision-making. Secondly, it also adds evidence on using financial literacy as a moderating variable and reports its significant impact on financial decision-making with other behavioral biases in Pakistan. Thirdly, the findings can help policymakers, investment banks/institutions, and stock markets to include the empirical evidence of the study to attract investors. They can also create various opportunities and activities to indulge the general public and investors to increase their intention to invest in various securities.

A limited number of studies in emerging/developing economies, especially in the context of Pakistan, have encouraged the need for current study. The impact of behavioral biases and financial literacy on individual investors' investing decisions, factors that have not been considered in Pakistan, is one of the novel foundations covered by this study. The findings will add to the empirical evidence on behavioral finance in the context of emerging financial markets of Pakistan, which has been overlooked in past studies. Practically, the study's findings will support investors in considering the significant behavioral factors and the importance of financial knowledge in making profit-oriented investment decisions.

The remaining portions of the study are divided into the following sections. The literature review is presented in segment two. Segment 3 provides a detailed description of the research methods utilized in the study, as well as the information sources and tests employed to assess the various hypotheses. Segment 4 presents the study's empirical findings and outcomes. The study's results and limitations are discussed in segment 5 to conclude the research.

2. Literature review

2.1. Individual investor's decision-making and behavioral biases

Baker, Kumar, Goyal, & Gaur, 2019 have identified several common behavioral biases among investors in behavioral psychology and finance. Behavioral biases frequently cause abnormalities in investors' decision-making. Scholars have explored how cognitive biases affect individual investor decision-making in various countries and communities. Ahmad & Shah, 2020 found that due to human psychology, investors make imperfect decisions. Behavioral biases, such as heuristics and herding, influence investment decisions while designing investment portfolios and other broader variables that affect investment decisions (Gavrilakis & Floros, 2021).

2.1.1. Overconfidence bias

Bhandari and Deaves (2006) stated that overconfidence bias is a person's propensity to overestimate their knowledge, skills, and the authenticity of their data. Overconfidence also explains the high trading

volumes in the markets. The studies by Pradikasari and Isbanah (2018) and Adil, Singh, & Ansari, 2021 found that it has a considerable detrimental impact on investor decision-making. According to Ahmad & Shah, 2020, overconfidence can decrease the ability of investors to make selection choices and performance. However, financial knowledge and risk perception can improve investors' decision-making.

2.1.2. Herding bias

Herding is used to describe the practice of investors following the trends and acknowledging the advice of others (Clarke et al., 2014). Rasool & Ullah, 2020 stated that the herding effect is a phenomenon where an investor's inclination is characterized as imitating the acts of others, which allows investors to avoid feeling guilty about their own decisions. According to Feng and Seasholes (2004) study on China, individual Chinese investors' herding behavior is significantly linked to the investment decision. Kumar & Goyal, 2016 and Gavrilakis & Floros, 2021 also reported the link between herd behavior and individual investors.

2.1.3. Disposition bias

Fogel and Berry (2010) defined disposition bias as an idea where Individuals regret selling a profitable investment too soon and keep holding onto a losing one for too long. Empirically, Baker, Kumar, Goyal, & Gaur, 2019 reported that investors are more inclined to hold losing stocks and likely to sell the appreciating/winning stocks. The studies by Kumar & Goyal, 2016, Mittal (2019), and Adil, Singh, & Ansari, 2021 also found the effect of disposition bias on investors' financial decisions. The Prosad, Kapoor, & Sengupta, 2015 also conclude that disposition bias significantly impacts decision-making.

2.1.4. Risk aversion

As per Tversky and Kahneman (1974), individuals naturally are risk averse and give more weight to losses than the same gain. They make their decisions in bonded rationality. According to Rasool & Ullah, 2020, under the assumption of portfolio theory, the investor would select the less risky investment among the available opportunities. The study by Maxfield, Shapiro, Gupta, & Hass, 2010 and Ball et al. (2010) concluded that risk aversion significantly impacts financial decision-making. They further reported, on a gender basis, that women are more inclined to be risk-averse than their male counterparts.

2.1.5. Representativeness bias

Busenitz and Barney (1997) explain representativeness bias as a phenomenon that results from a few observations, generally from small and non-random samples. Daniel et al. (1998), Khan, Afeef, Jan, & Ihsan, 2021, and Khan, Bashir, & Islam (2021) disclosed that it significantly and favorably affects investor decision-making and investors' overreaction in the market. The study by Ahmad, Shah, & Abbass, 2020 reported that representativeness, a heuristic bias, has a detrimental impact on the strategic decisions of entrepreneurs in global markets.

2.1.6. Anchoring and adjustment bias

According to Shah, Ahmad, & Mahmood, 2018, anchoring bias is another psychological heuristic highlighting people's propensity to base their judgments only on the first or most pertinent information they receive. The anchoring effect persists after a theoretical understanding of financial behavior (See. Palm & Andersson, 2020). In this bias, people produced predictions by initiating an original value and gradually increasing it to reach a final response, which is currently being investigated (Tversky & Kahneman, 1974; Baker, Kumar, Goyal, & Gaur, 2019;). Jain, Walia, & Gupta, 2019 discovered that anchoring bias significantly and positively, while Shah, Ahmad, & Mahmood, 2018 found negatively and substantially influences investors' decisions.

2.2. Financial literacy and investor decision-making

According to Noctor et al. (1992), "the capability of making well-informed judgments about utilizing and managing money using information and skills characterized as financial literacy. Empirically, Khan et al. (2024) and Silva et al. (2022) concluded that more financially educated people are more cautious about their investment decisions. Furthermore, Kasoga (2021) stated that financial literacy decreases cognitive biases and heuristics, significantly influencing investors' behavior in financial markets. Thus, there is a relationship between financial literacy and investment decisions. It further infers that bad investment selections/decisions are due to a lack of financial expertise.

2.3. Behavioral biases, investor decision-making, and financial literacy

Fernandes et al. (2014) reported that behavioral psychologists and financial academics have identified many behavioral biases related to investors. Baker, Kumar, Goyal, & Gaur, 2019 examined and reported an association between financial literacy, financial education, and financial behavior. Empirically, studies like Al-Tamimi (2009), Ateş et al. (2016), and Adil, Singh, & Ansari, 2021 reported that behavioral biases and financial literacy significantly influence investor behavior. Theoretically, behavioral finance emphasizes the relationship between financial education and various economic behavior traits (See. Ateş et al., 2016). It highlighted that comprehending individual investing decisions entails an awareness of the different behavioral biases that influence decision-making. Thus, it can be concluded that financial knowledge is necessary to understand investors' behavior, but empirical evidence is still inconclusive. In the existing empirical studies, financial literacy has not been identified as a moderator variable between behavior biases and investment decisions. Adil, Singh, & Ansari, 2021 stated that the moderator variable would impact the intensity or shape of the link in the dependent and independent variables and can be modified.

The literature review highlighted the mixed results and inconclusive relationship between various behavioral biases and investors' decision-making process. Therefore, based on the literature review, the study hypothesizes the following hypothesis to add empirical evidence of Pakistani investors to the behavioral finance field and address the existing gap.

2.4. Hypotheses development

H1. The overconfidence bias significantly influences the investment decisions of investors.

H2. The herding bias significantly influences the investment decisions of investors.

H3. Disposition bias significantly influences investors' investment decisions.

H4. The risk aversion bias significantly influences the investment decisions of investors.

H5. The representativeness bias significantly influences investors' investment decisions.

H6. The anchoring bias significantly influences the investment decisions of investors.

H7. Financial literacy significantly influences the investment decisions of investors.

H8. The relation between overconfidence bias and investors' decision-making is moderated by financial literacy.

H9. The relation between herding bias and investors' decision-making is moderated by financial literacy.

H10. The relation between disposition bias and investors' decision-

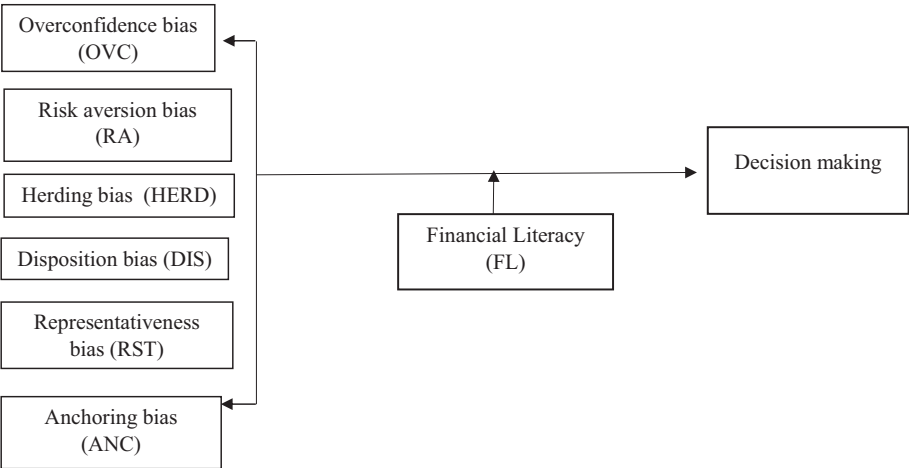
making is moderated by financial literacy.

H11. The relation between risk aversion and investors’ decision-making is moderated by financial literacy.

H12. The relation between representativeness bias and investors’ decision-making is moderated by financial literacy.

H13. The relation between anchoring bias and investors’ decision-making is moderated by financial literacy.

2.5. Research framework



Source: Authors’ compilation based on literature review.

3. Research methodology

3.1. Data collection and target population

This study uses primary data to test the hypothesis. Primary data is obtained for the first time for a specific purpose. The primary data closely resembles the behavioral features of investors while making investment decisions (Lin, 2011). The study used a cross-sectional research approach in which data includes individuals and periods. Individuals who engage in various financial opportunities as financial assets (Stock market, mutual funds, etc.) are considered responders in this research. The data was collected from various investors in Pakistan. Respondents were individually reached by traveling to each bank and financial institution to collect data. The adopted questionnaire was presented to them by hand for replies as it was filled out by email and online social media platforms (LinkedIn, Facebook, etc.). The department ethics committee reviewed and approved this study involving human participants. The data did not require the collection of personal information, and there were no issues of any disclosure or privacy. Therefore, this study did not require written informed consent for participation by national legislation and institutional requirements, so data is only from willing respondents. Three hundred questionnaires were delivered to different financial institutions, 261 responses were appropriately collected and used for analysis, and the remaining incomplete responses were excluded.

3.2. Sampling technique

The study used a convenient sampling approach, which is cheap and easy to implement. Convenience sampling is a non-probability sampling that selects only individuals from the intended audience who fulfill specified practical requirements, such as accessibility (Bell et al., 2022). This sampling approach collects data from a subset of the population that is readily available. In the current study, the respondents are individuals from Pakistan who invest in several investments. Pakistan is a developing economy with significant growth potential, and investigating investors’ attitudes could help to understand the behavior of investors.

3.3. Measurements of variables

This research questionnaire comprised thirty-eight questions for measuring behavioral biases, i.e., overconfidence, risk aversion, herding, disposition, representativeness, and anchoring. Section A and Section B are the two portions of the questionnaire. Section A consists of the respondents’ demographic profiles, which allow them to provide personal information, such as age, gender, education, and investment experience. Section B is scenario-based, requiring responders to identify connections to the artificial financial market situation. The scenario is designed so that their responses mirror their fundamental behavioral biases, financial literacy, and investing decisions. Financial literacy has two-part (Yes or No) questions called the dichotomous question (Adil, Singh, & Ansari, 2021; Lusardi & Mitchell, 2014). The type of questions and 5-point Likert scale model ranging from one (strongly agree) to five (strongly disagree) is adapted from existing studies such as (Adil, Singh, & Ansari, 2021; Lin, 2011; Lusardi & Mitchell, 2014). The measurements of variables such as behavioral biases (overconfidence, risk aversion, herding, and disposition), financial literacy, and investment decision-making are adopted from Adil, Singh, & Ansari, 2021, and the remaining two variables measurements (representative bias and anchoring) have been taken from Jain, Walia, & Gupta, 2019.

3.4. Statistical techniques for data analysis

Pilot testing was undertaken to ensure the instrument’s validity and reliability and to improve the questionnaire by eliminating low-quality items, such as incomplete data or biased ratings. The pilot study results indicated moving forward with the data collection process as no issue was found in it. The Pearson correlation and Cronbach’s alpha tests were utilized to evaluate the validity and reliability of the data. Moreover, the

reliability test results (Anchoring 0.733, Overconfidence 0.735, Disposition 0.670, Herding 0.708, risk aversion 0.737, Representativeness 0.746, investment decision 0.744, and Financial literacy 0.464) ensure the reliability of data. The hierarchical regression analysis adopted from the existing studies (Adil, Singh, & Ansari, 2021; Lin, 2011; Lusardi & Mitchell, 2014) was used in the study to assess the hypothesis. It follows a successive order where stepwise entry of the predictor variables takes place for analysis. This method is commonly used to examine the influence of the predictor variable while contorting the others (Adil, Singh, & Ansari, 2021; Lewis, 2007). A descriptive test is used to test the demographic variables. After that, moderation analysis was done by process macro to evaluate whether financial literacy moderates behavioral biases and investment decision-making, and a moderation impact between behavioral biases and financial literacy was found.

4. Empirical results and discussions

4.1. Pilot testing

The instrument’s reliability of the elements contained in it was determined through pilot testing. For this, 50 questionnaires were distributed by hand to individual investors for pilot testing. Only 35 surveys were received, 30 being used, as the remaining were incomplete.

4.2. Reliability test

The reliability of the variables is assessed using Cronbach’s alpha coefficient method. Cronbach’s alpha is a statistic that quantifies the scale’s inter-item consistency of numerous items (Baker, Kumar, Goyal, & Gaur, 2019). Cronbach’s alpha reliability testing determines if two or more things are related. According to research, instruments must have a reliability coefficient of 0.5 to 0.8 to create consistency (Adil, Singh, & Ansari, 2021; Pedhazur, 1982).

The behavioral biases score in this study is reliable, according to Cronbach’s alpha test, as anchoring bias 0.733, overconfidence bias 0.735, disposition bias 0.670, herding bias 0.708, risk aversion 0.737, representativeness bias 0.746, investment decision 0.744, and financial literacy is 0.464. The result of the current study is consistent with (Adil, Singh, & Ansari, 2021; Baker, Kumar, Goyal, & Gaur, 2019; Rasool & Ullah, 2020). Table 1 displays the overall results of the reliability and validity test. The findings indicate that the variables have a more substantial level of internal consistency. These indices suggest that the measurements in the variables are trustworthy and can proceed further analysis.

4.3. Descriptive test

Based on 261 complete responses, 210 were males, and 51 were females, 80.5 % and 19.5 %, respectively. In total, 20.5 % of the total respondents belong to the age group of 18–24, most of the respondents were found in the age of 25–35 which is 44.8 %, and the remaining respondents fell in the age group of 36–45 (27.6 %) and above 45 were only found 6.9 %. In the education background, most of the respondents

131 were postgraduate degree holders, 50.2 %. The remaining respondents were found to have graduated 89 (34.1 %). Following that, 23 respondents were undergraduates 8.8 %, and doctorate respondents were only 18 (6.9 %). According to descriptive statistics, 86 (33.0 %) individual investors having 4 to 5 years of investment experience, 74 (28.4 %) individual investors had more than five years of investment experience, 75 (28.7 %) investors had less than or one-year investment experience, and only 26 (10 %) investors having between 1- and 3-years investment experience. Fig. 1 depicts the key statistics about the demographic profile of the respondents.

4.4. Correlation analysis

Correlation analysis is used to determine the association between all the variables to check for multicollinearity. When the coefficient of correlation is small, there is no multicollinearity issue (Khan, Afeef, Jan, & Ihsan, 2021; Khan, Bashir, & Islam, 2021; Adil, Singh, & Ansari, 2021; and Khan, 2022). The results of correlations are reported in Table 2. The correlation coefficient is one ($r = 1$), and the findings reveal that each variable is entirely associated with itself. The results are the following (Shah, Ahmad, & Mahmood, 2018). According to the correlation analysis results, anchoring bias positively correlates with investment decision-making with a Pearson correlation ($r = 0.298$), which means anchoring bias is linked with investment decision-making. According to findings, overconfidence bias positively relates to disposition bias with Pearson correlation ($r = 0.020$), herding bias ($r = 0.192$), risk aversion ($r = 0.319$), and representativeness ($r = 0.273$), which means that due to an increase in overconfidence bias, disposition bias, representativeness bias, anchoring and adjustment, and herding bias also increase. The results are consistent with those (Adil, Singh, & Ansari, 2021; Shah, Ahmad, & Mahmood, 2018). According to psychology, representativeness, availability, anchoring, and adaptability also increase as overconfidence does (Shah, Ahmad, & Mahmood, 2018). The result also shows that financial literacy and overconfidence bias negatively correlate with the correlation result ($r = -0.076$).

As financial literacy increases, the overconfidence bias decreases. The findings indicate that risk aversion bias is negatively associated with financial literacy with Pearson correlation analysis ($r = -0.058$). Results indicate that representativeness bias is also negatively linked with financial literacy with a Pearson correlation result of ($r = -0.064$), which shows that due to an increase in the financial knowledge of investors; these biases (overconfidence, representativeness, and risk aversion bias) reduce. These findings are consistent with Adil, Singh, & Ansari, 2021, which show that behavioral biases negatively correlated with the moderator variable (financial literacy) and moderate the association between individual investors’ behavioral biases and decision-making, concluding that financial literacy significantly reduces behavioral biases.

4.5. Regression analysis

The hypotheses were tested using the regression analysis results given in Table 3. Behavioral biases are used as the primary predictor, and behavioral biases are anchoring bias, overconfidence, disposition, herding, risk aversion, and representativeness. The changes were entered in R2 and reported beta. The R2 change value is 0.205, $P = 0.000$, indicating that a 20.5 % change in the investment decision-making behavior of investors caused by overconfidence, anchoring, disposition, risk aversion, representativeness, and herding shows that there is a statistically significant association between individual investors behavioral biases and decision making. The value of F statistics ($F = 10.925$, sig 0.000) indicates that the model is fit and statistically significant. Moreover, findings suggest that the behavioral biases as anchoring bias (Beta = 0.265, P -value <0.05), overconfidence bias (Beta = 0.125, P -value <0.05), Herding (Beta = 0.246, P -value <0.05) are statistically significant to the dependent variable investment

Table 1
Reliability test results.

Variables	Cronbach Alpha reliability test	No of item
Anchoring Bias	0.733	4
Overconfidence bias	0.735	5
Disposition bias	0.670	5
Herding bias	0.708	5
Risk Aversion	0.737	5
Representativeness	0.746	5
Investment Decision	0.744	5
Financial Literacy	0.464	4

Source: Authors’ compilation.

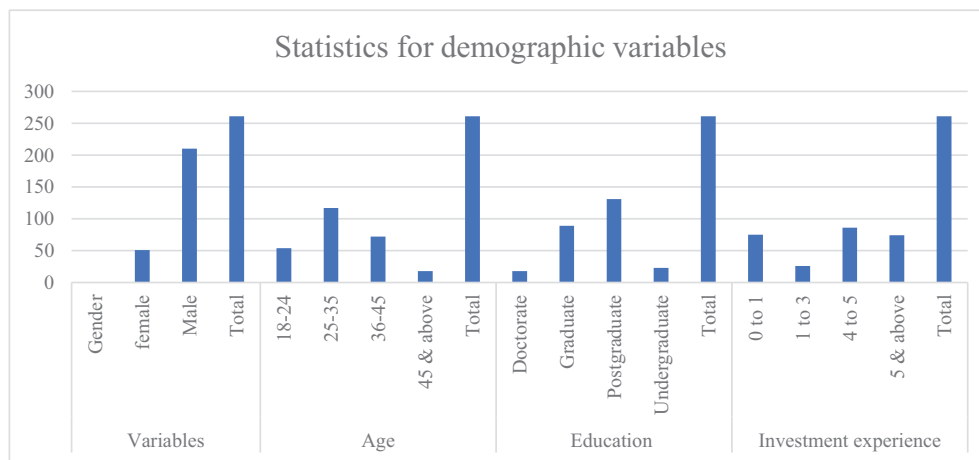


Fig. 1. Statistics for demographic variables.

Source: Authors' compilation.

Table 2

Pearson correlation analysis.

	Mean	SD	Anchoring bias	Overconfidence	Disposition	Herding	Risk aversion	Representativeness	Investment decision	Financial literacy
Anchoring Bias	9.80	3.91	1							
Overconfidence	10.67	4.43	−0.125*	1						
Disposition	11.21	4.11	0.245**	0.020	1					
Herding	10.86	4.08	0.216**	0.192**	0.278**	1				
Risk Aversion	10.75	4.65	0.070	0.319**	0.283**	0.251**	1			
Representativeness	11.12	4.53	0.140*	0.273**	0.270**	0.202**	0.392**	1		
Investment Decision	10.96	4.62	0.298**	0.180**	0.100	0.338**	0.200**	0.196**	1	
Financial Literacy	7.68	0.66	0.055	−0.076	0.029	−0.039	−0.053	−0.058	−0.064	1

* Correlation is significant at the 0.05 level (2-tailed).

** Correlation is significant at the 0.01 level (2-tailed).

Source: Authors' compilation.

Table 3

Regression test score, variables in the equation:

Coefficients ^a						
Model		Unstandardized coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	2.879	1.189		2.421	0.016
	AnchoringBias	0.313	0.071	0.265	4.434	0.000
	Overconfidence	0.131	0.064	0.125	2.028	0.044
	Disposition	−0.085	0.070	−0.076	−1.219	0.224
	Herding	0.278	0.069	0.246	4.031	0.000
	RiskAversion	0.074	0.064	0.075	1.160	0.247
	Representativeness	0.068	0.065	0.066	1.048	0.296
	Financial literacy	−0.445	0.428	−0.064	−1.039	0.300

^a Dependent Variable: InvestmentDecision.

Source: Authors' compilation.

decision making.

Furthermore, risk aversion (Beta = 0.075, P-value > 0.05) and representativeness bias (Beta = 0.066, P-value > 0.05) are statistically insignificant, and disposition bias (Beta = −0.076, P-value > 0.05) are statistically and negatively insignificant to dependent variables. The unstandardized coefficient beta shows that all the behavioral biases positively and significantly affect investment decisions except disposition, which has a negative association. The unstandardized coefficient only reports the association between dependent and independent variables, not among the independent variables (Bhalla, 2015). This study's results are consistent with the latest research (Adil, Singh, & Ansari, 2021), which indicates that behavioral biases such as overconfidence

bias and herding significantly impact investment decision-making, and disposition bias is insignificantly linked with investment decision-making; these results are similar to Ahmad (2021). It suggests that psychological biases (representativeness, anchoring, herding, and overconfidence) affected Malaysian institutional and individual investors.

The findings also indicate that financial literacy (B = −0.445, P > 0.30) has negatively impacted investors' decision-making. It could be interpreted in two ways. Firstly, from the results, anchoring and herding have a higher propensity to affect decisions, which is more psychological than logical. Secondly, financial literacy, which may help analyze the investment logically, may reduce the investors' intention to invest. This

Table 4
Moderation effect of financial literacy.

Predictor	Beta	Moderation	Significance level
Anchoring Bias	0.265	ANC × FL	0.39
Overconfidence	0.125	OVC × FL	0.25
Disposition	−0.076	DIS × FL	0.35
Herding	0.246	HERD × FL	0.25
Risk Aversion	0.075	RV × FL	0.23
Representativeness	0.066	RST × FL	0.34

Source: Authors' compilation.

means that financial knowledge and behavioral biases affect the proportion of investors' investments (Adil, Singh, & Ansari, 2021). The estimation results support the hypotheses (H1, H2, and H6), i.e., only overconfidence, herding, and anchoring bias influence the investment decisions of investors in Pakistan. In contrast, the study's findings do not support other hypotheses (H3, H4, H5, and H7). This fractional association with prospect theory proposes that psychological traits like overconfidence and herding may supersede rational decision-making processes, forcing investors to depart from purely rational choices prescribed by traditional finance theories

Bashir et al.'s (2024) study on China reported empirically similar investor sentiment results. However, with more financial knowledge, rationality prevails over psychological traits, as reported by Khan et al. (2024); while in the absence of knowledge, behavioral biases significantly influence the investment decisions of Pakistani investors.

4.6. Moderation analysis

Finally, how financial literacy moderates the relationship between individual investors' behavioral biases and investment decision-making has been tested. We examine the moderation effect through Process macro by entering six independent variables individually (anchoring bias, overconfidence bias, herding bias, disposition bias, risk aversion, and representativeness bias) with individual investors' dependent variable investment decision-making and moderation impact of financial literacy. Process macro explores the impact of the independent variable on the dependent variable through the moderating variable (Hayes, 2017). The findings of the moderation analysis are reported in Table 4.

According to the results, behavioral biases such as anchoring, overconfidence, herding, risk aversion, and representativeness ($B = 0.265$, $Sig = 0.39$), ($B = 0.125$, $Sig = 0.25$), ($B = 0.246$, $Sig = 0.25$), ($B = 0.075$, $Sig = 0.23$), ($B = 0.066$, $Sig = 0.34$) were found positive and the statistically insignificant moderating role of financial literacy between behavioral biases and investment decision making. Moreover, the interaction with disposition bias ($B = -0.076$, $Sig = 0.35$) was insignificant. The study findings are consistent with (Adil, Singh, & Ansari, 2021), who concluded that the interaction of financial literacy with behavioral biases such as disposition, herding, and risk aversion was statistically insignificant. Furthermore, the moderation analysis results reject the hypotheses (H8, H9, H10, H11, H12, H13).

The current investigation results are similar (Ahmad, 2021; Baker, Kumar, Goyal, & Gaur, 2019), suggesting that psychological biases, including representativeness bias, anchoring bias, herding bias, and overconfidence bias, statistically considerably influence investors' decision-making. Due to these subjective impressions, these biases cause investors to make poor decisions despite the availability of relevant information. In order to improve investment strategies and overall financial performance, it is imperative to acknowledge and mitigate these biases. The findings of the study are similar to the findings of Roth and Voskort (2014), Shah, Ahmad, & Mahmood, 2018, Trönnberg & Hemlin, 2019, Özen and Ersoy (2019), Rasool & Ullah, 2020, Gavrilakis & Floros, 2021, Kasoga (2021), and Jain, Walia, Kaur, & Singh, 2021 they also reported that distinct behavioral biases statistically influence the individual investors' investment decision-making. The study by Mushinada & Veluri, 2019 also stated that behavioral biases like

overconfidence and self-attribution bias significantly impact investors' investment choices. Hence, it can be inferred that the investor's confidence level increases with his/her education level and investment experience.

The results also suggest that male investors are more overconfident than female investors in Pakistan. Financial literacy insignificantly moderates the relationship between behavioral biases and investment decision-making; this aligns with the findings of Son and Park (2019) and Adil, Singh, & Ansari, 2021. These studies suggested that insignificance is a lack of financial knowledge, which could lead to irrational investment decisions. To overcome this problem, they proposed a higher level of financial knowledge for investments in profitable investments. Likewise, Ramalho and Forte (2019) and Baker, Kumar, Goyal, & Gaur, 2019 supported the argument that financial knowledge could help investors to overcome the mentioned behavioral biases. Therefore, these results indicate that enhancing individual investors' financial knowledge and skills is essential for making fruitful investment decisions in competitive financial instruments. Consequently, a high level of financial literacy could notably empower investors to make well-informed and rational decisions, ultimately leading to improved financial outcomes.

In summary, the study's conclusions have the following practical ramifications. First, to improve investment decisions, individual investors must comprehend and consider the behavioral biases that were analyzed. Second, education, counseling, and tailored investing solutions can help financial institutions, brokers, and advisors overcome these behavioral biases. Investors can make more informed and logical judgments if they are aware of and account for these biases. In addition, legislators should consider incorporating behavioral finance concepts into regulatory frameworks to encourage openness and safeguard investors' investments.

5. Conclusion and recommendations

This research empirically investigated the influence of behavioral biases (anchoring bias, risk aversion bias, herding bias, representativeness bias, overconfidence bias, and disposition bias) on individual investors' investing decisions and identified the moderating impact of behavioral biases and investors' decision-making.

We established a positive and significant association between behavioral biases (overconfidence, herding, and anchoring) and investment decision-making. Risk aversion, disposition, and representativeness bias are found statistically insignificant. This insignificance means that behavioral factors influence individual investors' decision-making while investors select investment securities. Moreover, it examined that financial literacy significantly moderates the relation between behavioral biases and investors' decision-making and insignificantly affects investors' decision-making. The research used an adopted questionnaire to collect data from individual investors. The Cronbach alpha test was used to evaluate the validity of variables.

Regression models and correlation tests were applied to test the hypothesis and analyze the relationship between variables. The findings indicate that behavioral biases strongly influence investment decision-making, and investors are not always behaving rationally, which is partially in line with the prediction of prospect theory, which indicates that investors have behavioral and psychological biases that impact their decision-making. This could be due to the fact that investors in developing countries do not want to spend more resources and time acquiring financial knowledge. If they acquire those skills and required knowledge, they may behave rationally. Otherwise, they may make investment decisions by following their behavioral traits or herd behavior. Investors must grasp these behavioral finance theories to restrain them from making irrational decisions. Behavioral finance clearly shows how investors make their decisions in reality.

A comprehensive study on behavioral finance can cover some significant behavioral biases that suggest that investors' biases

considerably influence their financial decisions. Investors make irrational decisions due to their mental shortcuts and biases. To overcome these biases, authors consider knowledge about financial products essential. People's capacity to acquire and utilize financial information expertly and confidently should be reflected in their financial decisions. Investors might take advantage to make better financial decisions if they are aware of behavioral biases and knowledge about financial products. Findings indicate that financial literacy is critical in overcoming these behavioral biases. It is suggested that a lack of technical competence would result from behavioral biases; however, an investor may make an appropriate decision based on their capabilities. Individual investors consult agents, brokers, and fund managers for advice or recommendations who have a deeper understanding of the financial market and may help them make a rational decision. This study further suggests that financial market courses be made available to investors to assist them in performing technical market analysis before investing. Further, it concludes that financial literacy is the remedy for overcoming investors' behavioral biases when making investment selections.

6. Implications

Based on the current research findings, several implications for management and individual investors might be made. They may improve their outcomes by acknowledging the biases and mistakes in judgment that we all make. Financial educators can use the findings to develop personal financial awareness initiatives, improve financial education, and implement other policies to boost individual financial talents and lead to economic prosperity that benefits both people and the economy. The research is also helpful in improving financial consultants' awareness of their clients' psychology, assisting businesses in creating a behaviorally adapted portfolio, helping investment bankers understand market sentiments, and helping financial strategists and security specialists make more accurate projections and stock recommendations. Furthermore, financial intermediaries and policymakers will benefit from these insights since they demonstrate that increasing people's financial literacy may considerably influence their financial decisions. Investors have been cautioned of the disastrous consequences of neglecting behavioral biases in investing analysis (Montier & Strategy, 2002). Therefore, the current study provides an understanding and comprehension of behavioral biases in capital management, which could be helpful for Pakistan's stock market investors.

7. Limitations and future directions

Despite the study's useful findings, there are several limitations to be aware of for future researchers. Firstly, the study covers the three most advanced and economically stable cities. Whereas the sample is from the whole country, investors with different educational, cultural, and economic circumstances and an increased population sample are desired in future studies. The sample description of this research is limited to only demographic factors of investors; we recommend that future researchers add more characteristics of investors to their portfolios and investment goals. Additionally, including factors such as access to financial markets for investors across the countries, costs related to these services could also impact their decision, which needs to be considered in future studies. Finally, other biases (availability bias, mental accounting bias, self-attribution bias, and conservatism bias) could be used in future studies.

CRedit authorship contribution statement

Faiq Mahmood: Supervision, Validation. **Rabia Arshad:** Data curation, Formal analysis, Writing – original draft. **Shoaib Khan:** Methodology, Validation. **Alia Afzal:** Data curation, Funding acquisition. **Mohsin Bashir:** Conceptualization.

Declaration of competing interest

We declare that we have no conflicts of interest that could influence the research conducted or the interpretation of the results presented in this manuscript. We have not received any financial support or funding from any organization that could have influenced the outcome of this study.

Furthermore, we confirm that this manuscript is our original work, has not been published previously, and is not under consideration for publication elsewhere. All authors have contributed significantly to the research, writing, and revision of this article and have approved the final version for submission.

Data availability

Data will be made available on request.

Acknowledgement

Authors are thankful to managing editor Dr. Cameron Peers and anonymous reviewers for their efforts and constructive comments to improve the overall quality of this manuscript.

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